

WASP-12b

R-0414

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230105 · created 2026-07-19T06:44:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459949.7007 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.087 +/- 0.03
Transit depth (Rp/Rs)^2	0.76 +/- 0.52 [%]
Orbital Inclination (inc)	84.67 +/- 2.85
Airmass coefficient 1 (a1)	1.001 +/- 0.018
Airmass coefficient 2 (a2)	-0.001 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	#2 - [86, 187]
Optimal Method	PSF photometry
Transit Duration (day)	0.1218 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.087 $\pm$ 0.03 vs 0.1178 (deviation 1.02 σ)
Duration fit vs published	0.1218 d vs 0.125 d (deviation 0.48 σ)
Mid-time O-C	-3.2 min
Depth SNR	2.9
Residual scatter	1.77 %

Light curve

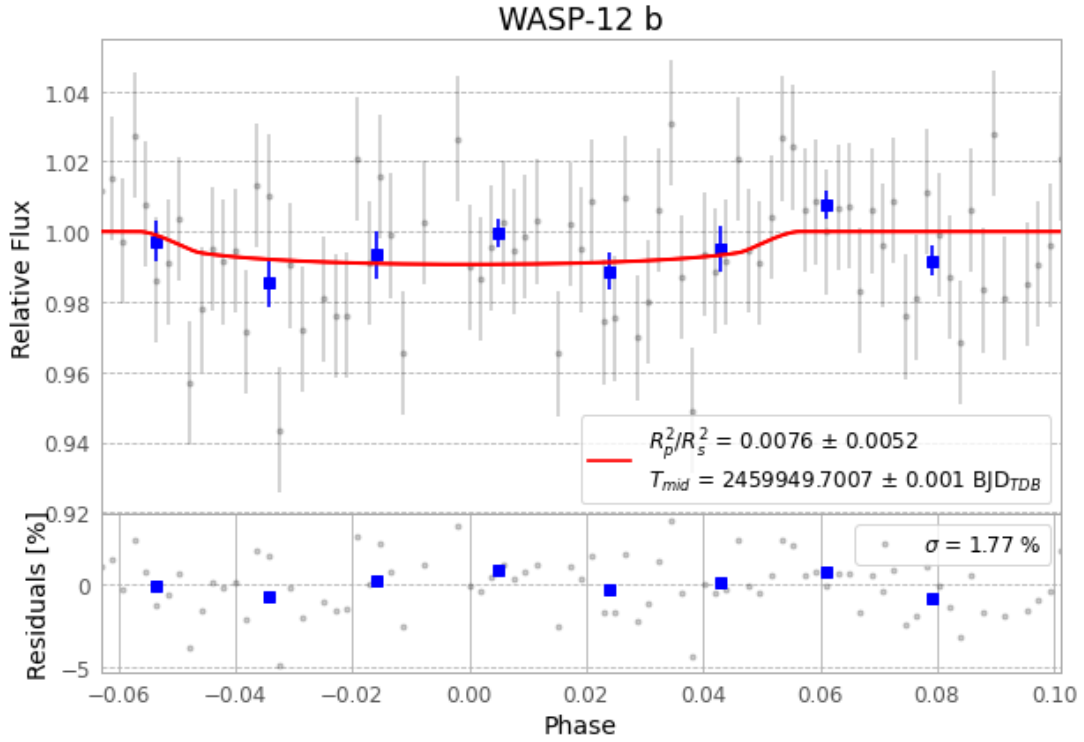


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-04.png (SHA-256 a6798523faa477b2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [428, 84], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 153e1c9e3c9b9a7b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

87 FITS frames (57 MB), WASP-12230105030015.FITS → WASP-12230105071815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230105.log (24e664aa5f0d...), FinalLightCurve_WASP-12 b_2023-01-04.png (a6798523faa4...), FinalLightCurve_WASP-12 b_2023-01-04.csv (8dd39160b48c...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 06:44Z.